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Pulsed-coherent electronic front end for lidar and radar detection and ranging

Abstract

Systems and methods for a light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF) measurement are disclosed. In one embodiment, a lidar system includes a reference clock providing a clock signal (CK.sub.ref) with time period T.sub.clk, an automatic gain control (AGC) loop that is triggered when a received signal RF.sub.in is greater than a threshold voltage V.sub.th, a coherent detector measuring a fine ToF by detecting the phase difference ($\Delta\phi$) between the clock signal (CK.sub.ref) and the received signal (RF.sub.in), a pulse edge detector measuring a coarse ToF by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$, and a combiner that calculates total ToF by combining output of the coherent detector and pulse edge detector using the equation: $T_{\text{ToF}} = [N + (\frac{\Delta\phi}{2\pi})] \times T_{\text{clk}}$.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7202941	12/2006	Munro	356/5.1	G01S 17/10
7388655	12/2007	Mori	356/5.1	G01S 7/497
8558993	12/2012	Newbury et al.	N/A	N/A
9515859	12/2015	McCormack	N/A	H01L 23/49861
10211592	12/2018	Villeneuve et al.	N/A	N/A
10732264	12/2019	Bailey	N/A	G01S 7/4816
11255663	12/2021	Binder	N/A	G01B 15/00
11722141	12/2022	Filippini	327/158	H03L 7/0812
11726184	12/2022	Ferreira	356/4.01	G01S 7/4815
11906290	12/2023	Binder	N/A	G01B 11/26
2004/0135992	12/2003	Munro	356/4.01	G01S 7/483
2008/0088818	12/2007	Mori	N/A	N/A
2008/0100822	12/2007	Munro	356/28	G01S 7/483
2011/0285980	12/2010	Newbury	356/3	G01S 7/484
2014/0355700	12/2013	Mccormack et al.	N/A	N/A
2019/0154439	12/2018	Binder	N/A	G01S 15/08
2019/0170859	12/2018	Bailey et al.	N/A	N/A
2020/0284883	12/2019	Ferreira	N/A	G01S 17/10
2022/0128352	12/2021	Binder	N/A	G01B 15/00
2024/0159522	12/2023	Binder	N/A	G01B 11/26
2024/0175678	12/2023	Binder	N/A	G01S 15/42

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2021007454	12/2020	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability for International Application

PCT/US2020/041443, Report issued Jan. 11, 2022, Mailed on Jan. 20, 2022, 06 Pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/US2020/041443, Search completed Sep. 21, 2020, Mailed Oct. 9, 2020, 14 Pgs. cited by applicant

Andersen et al., “A 118-mW 23.3-GS/s dual-band 7.3-GHz and 8.7-GHz impulse-based direct RF sampling radar SoC in 55-nm CMOS”, 2017 IEEE International Solid-State Circuits Conference (ISSCC), Feb. 5-9, 2017, San Francisco, CA, USA, pp. 138-139, DOI:

10.1109/ISSCC.2017.7870299. cited by applicant

Behroozpour et al., “Chip-scale electro-optical 3D Fmcw lidar with 8μm ranging precision”, 2016 IEEE International Solid-State Circuits Conference (ISSCC), Jan. 31-Feb. 4, 2016, San Francisco, CA, USA, pp. 214-216, DOI: 10.1109/ISSCC.2016.7417983. cited by applicant

Chen et al., “A 19-GHz Pulsed-Coherent ToF Receiver With 40-μm Precision for Laser Ranging Systems”, In ESSCIRC 2019-IEEE 45th European Solid State Circuits Conference (ESSCIRC), Sep. 23, 2019, pp. 191-194 [online], [retrieved on Sep. 11, 2020]. Retrieved from the Internet

<URL: <https://ieeexplore.IEEE.org/abstract/document/8902597>>, entire document. cited by applicant

Ellinger et al., “Analysis and Compensation of Phase Variations Versus Gain in Amplifiers Verified by SiGe HBT Cascode RFIC”, IEEE Transactions on Microwave Theory and Techniques, Aug. 2009, First Published: Jul. 6, 2009, vol. 57, No. 8, pp. 1885-1894, DOI:

10.1109/TMTT.2009.2025415. cited by applicant

Hafez et al., “Analysis and Design of Superharmonic Injection-Locked Multipath Ring Oscillators”, IEEE Transactions on Circuits and Systems I: Regular Papers, Jul. 2013, First Published: Jan. 25, 2013, vol. 60, No. 7, pp. 1712-1725, DOI: 10.1109/TCSI.2012.2230591. cited by applicant

Huang et al., “Mode-Locked Ultrashort Pulse Generation from On-Chip Normal Dispersion Microresonators”, Physical Review Letters, Feb. 6, 2015, vol. 114, p. 053901-1-053901-5, doi: 10.1103/PhysRevLett.114.053901. cited by applicant

Kim et al., “A High Noise Immunity, 28 X 16-Channel Finger Touch Sensing IC Using OFDM and Frequency Translation Technique”, Sensors, vol. 18, No. 5, May 21, 2018, p. 1652 (online), [retrieved on Sep. 11, 2020]. Retrieved from the Internet <URL: <https://www.mdpi.com/1424-8220/18/5/1652>>, entire document. cited by applicant

Nissinen et al., “Integrated Receiver Including Both Receiver Channel and TDC for a Pulsed Time-of-Flight Laser Rangefinder With cm-Level Accuracy”, IEEE Journal of Solid-State Circuits, May 2009, vol. 44, No. 5, pp. 1486-1497, DOI: 10.1109/JSSC.2009.2017006. cited by applicant

Payne et al., “A 512×424 CMOS 3D Time-of-Flight image sensor with multi-frequency photo-demodulation up to 130MHz and 2GS/s ADC”, 2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC), Feb. 9-13, 2014, San Francisco, CA, USA, pp. 134-135, DOI: 10.1109/ISSCC.2014.6757370. cited by applicant

Yasutomi et al., “A 0.3mm-resolution Time-of-Flight CMOS range imager with column-gating clock-skew calibration”, 2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC), Feb. 9-13, 2014, San Francisco, CA, USA, pp. 132-133, DOI: 10.1109/ISSCC.2014.6757369. cited by applicant

Zhu et al., “Synthetic-wavelength-based dual-comb interferometry for fast and precise absolute

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The current application is a national stage of PCT Patent Application No. PCT/US2020/041443 entitled “Pulsed-Coherent Electronic Front End for Lidar and Radar Detection and Ranging” filed Jul. 9, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/872,126 entitled “Pulsed-Coherent Electronic Front End for Detection and Ranging” filed Jul. 9, 2019, the disclosures of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Lidar (Light Detection and Ranging) and radar are methods for distance ranging by measuring the time-of-flight (ToF) of light or radio-waves. In lidar, the target is typically illuminated with laser light and the reflection is measured with a sensor. Differences in laser return times and wave characteristics can then be used to construct a distance and/or location of the target. Modern applications of lidar and radar systems include self-driving automobiles and autonomous robots.

(2) Three main detection methods are used in ranging systems. The first is coherent detection (also known as interferometry), which can achieve high resolution but has slow acquisition due to the range ambiguity. Coherent detection generally measures changes in phase of the reflected light. The second, frequency-modulated continuous-wave (FMCW) detection, sets up a laser to emit linear optical frequency chirps. This has both moderate resolution and acquisition but narrow dynamic range. The third and the last, pulsed detection, emits short pulses or pulse patterns from the sensor aperture and the power of back-reflected light is detected using a square-law photodetector. This has low resolution limited by timing offsets commonly referred to as “walk error”, but it has the advantages of fast acquisition and long-distance measurement. Demand for ranging systems with high depth resolution and fast acquisition has emerged for various applications such as mobile 3D scanning and remote biometrics, but, as state-of-the-art systems show (FIG. 1), simultaneously satisfying both requirements can be challenging.

SUMMARY OF THE INVENTION

(3) Systems and methods for a light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF) measurement are disclosed. In one embodiment, a lidar system includes a reference clock providing a clock signal (CK.sub.ref) with time period T.sub.clk, an automatic gain control (AGC) loop that is triggered when a received signal RF.sub.in is greater than a threshold voltage V.sub.th, a coherent detector configured to measure a fine ToF by detecting the phase difference ($\Delta\phi$) between the clock signal (CK.sub.ref) and the received signal (RF.sub.in), a pulse edge detector configured to measure a coarse ToF by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$, a combiner configured to calculate total ToF by combining output of the coherent detector and pulse edge detector using the equation,

$$(4) \text{ } T o F = [N + (\frac{\Delta\phi}{2\pi})] \times T_{clk} .$$

(5) In another embodiment, the AGC loop includes a folded-cascode amplifier as a V/I converter.

(6) In a further embodiment, the coherent detector includes two single-side band (SSB) mixers.

(7) A still another embodiment includes a variable gain analog front-end to control amplitude of the

received signal (RF.sub.in).

(8) In a yet further embodiment, the variable gain analog front-end comprises a phase-invariant variable-gain low-noise amplifier (PI-VGLNA), in-phase and quadrature phase (I/Q) down-conversion mixer, programmable gain amplifier (PGA) and variable gain amplifier (VGA).

(9) In another further embodiment, the PI-VGLNA comprises a current-steering cascode architecture with inductors between common source and common gain stages.

(10) In an additional embodiment, the PGA and VGA comprise current-steering structures controlled by single-to-differential V/I converters.

(11) In another additional embodiment, the pulse edge detector comprises varactors adjusted according to voltage V.sub.ctrl of the AGC loop, 8-way time-interleaved samplers, an 8-to-16 demultiplexer, and XOR gates.

(12) In a still further embodiment, a method for measuring distance with a light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF) includes providing a clock signal (CK.sub.ref) with time period T.sub.clk from a reference clock, triggering an automatic gain control (AGC) loop when a received signal RF.sub.in is greater than a threshold voltage V.sub.th, measuring a fine ToF using a coherent detector by detecting the phase difference ($\Delta\phi$) between the clock signal (CK.sub.ref) and the received signal (RF.sub.in), measuring a coarse ToF using a pulse edge detector by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$, and calculating total ToF by combining output of the coherent detector and pulse edge detector using the equation:

(13) $\text{ToF} = [N(\frac{1}{2})] \times T_{\text{clk}}$.

(14) Another embodiment also includes controlling amplitude of the received signal (RF.sub.in) using a variable gain analog front-end.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 graphically illustrates capabilities of different types of ranging detection in resolution vs. sampling rate.

(2) FIG. 2A illustrates an example waveform that may be utilized for measurements in pulsed-coherent range detection in accordance with several embodiments of the invention.

(3) FIG. 2B is a circuit diagram illustrating a pulsed-coherent range detector in accordance with several embodiments of the invention.

(4) FIG. 3 is a circuit diagram illustrating a receiver architecture for pulsed-coherent range detection in accordance with several embodiments of the invention.

(5) FIG. 4A is a circuit diagram illustrating a phase-invariant variable-gain low-noise amplifier (PI-VGLNA) in accordance with several embodiments of the invention.

(6) FIG. 4B is a circuit diagram illustrating a programmable gain amplifier/variable gain amplifier (PGA/VGA) in accordance with several embodiments of the invention.

(7) FIG. 5 is a circuit diagram illustrating a coarse time of flight (ToF) detector in accordance with several embodiments of the invention.

(8) FIG. 6 is a circuit diagram illustrating a superharmonic injection locked multipath ring oscillator (SHIL-MPRO) providing 8-way interleaving clock phases in accordance with several embodiments of the invention.

(9) FIG. 7 is a circuit diagram illustrating a measurement system for receivers in accordance with embodiments of the invention.

(10) FIG. 8 graphically illustrates example measurement results of a phase-invariant AFE.

(11) FIGS. 9A and 9B graphically illustrate an AGC loop characteristic in certain situations.

(12) FIGS. 9C and 9D compare walk error in certain situations.

(13) FIG. 10A graphically illustrates rms error of a coarse ToF measurement in certain situations.

(14) FIG. 10B graphically illustrates rms error of a fine ToF measurement in certain situations.

(15) FIG. 11 graphically illustrates an example spectrum of a received pulse sequence at the VGA's output in accordance with an embodiment of the invention.

(16) FIG. 12 graphically illustrates phase noise of a 4.75-GHz clock for coherent measurement in accordance with an embodiment of the invention.

(17) FIG. 13 illustrates a process for measuring distance using both coherent and pulsed ToF detection in accordance with an embodiment of the invention.

DETAILED DISCLOSURE OF THE INVENTION

(18) Turning now to the drawings, pulsed-coherent electronic front ends for lidar and radar detection and ranging in accordance with embodiments of the invention are disclosed. For conventional coherent detection to achieve sub-mm resolution with 10-m dynamic range without ambiguity (covered within one period of the signal), it requires 17-bits of resolution for the analog-to-digital converter (ADC) or time-to-digital converter (TDC) to acquire phase information. Segmented measurements with frequency sweeping or dual frequency combs can be used to ease this requirement on the ADC. The acquisition is usually slow due to the scanning behavior, and furthermore, the search time for the coarse measurement increases dramatically for long-distance ranging.

(19) In many embodiments of the invention, pulsed-coherent segmented time-of-flight (ToF) measurement is utilized to enhance the sampling rate, which leverages the high resolution of coherent detection and the fast acquisition of pulsed detection. Pulsed detector circuitry may be utilized for a coarse ToF measurement, while coherent detector circuitry may be utilized for a fine ToF measurement. The coarse and fine ToF measurements can then be combined to generate a total ToF measurement.

(20) In one embodiment, a pulsed-coherent lidar system is designed with a 19-GHz carrier frequency which fits with a 19-GHz repetition rate ultra-low jitter mode-locked laser (MLL) to enable further lidar system integration. Using a high carrier frequency eases the requirements of the ADC's resolution for coherent detection. A 9-bit medium resolution ADC is sufficient for resolution in the tens of microns with a 19-GHz carrier. To break the tradeoff between the precision and the acquisition rates, this 19-GHz carrier can be modulated with 6.8-ns pulses, and the arrival time of the pulse's envelope can be measured by a 19-GHz counter as the coarse measurement. This approach also provides a capability of high dynamic measurement range. With sufficient sensitivity of the front-end, the acquisition range can be linearly increased by adding more bits to the counter. There can be two challenges with this pulsed-coherent approach that are each addressed below. First, the walk error of the envelope pulse detection can easily be larger than a 19-GHz period, thus limiting the resolution. Second, linearity is important in a coherent receiver to maintain fine phase information. Using a variable gain amplifier (VGA) can increase the dynamic range, but phase variation may be introduced. As will be described below in accordance with certain embodiments of the invention, post-edge pulsed detection can be utilized to suppress walk error within a clock period, and a phase-invariant variable gain analog front-end (AFE) to minimize the phase variation. One skilled in the art will recognize that characteristics, such as carrier frequencies and pulse durations, of a system or method performing such measurements may be different in additional embodiments as appropriate to a particular application.

(21) The disclosure of the present application includes at least three main concepts. First, combining pulsed based detection and ranging commonly used in long distance LiDARs with coherent detection is often used for very high measurement accuracy over short distances. This approach enables one to dynamically tradeoff the resolution and sampling rate (of a LiDAR measurement) benefitting from the multiple measurement and improved noise of coherent detection (for fine measurement) and the speed of the pulsed detection (for coarse measurement).

(22) Second, walk error is a well-known problem that introduces detection error for varying input signal amplitude. This issue can be addressed with the use of a pulsed-coherent architecture in accordance with embodiments of the invention. When the pulse is used only for coarse signal detection, a ranging system in accordance with several embodiments can tolerate a larger amount of error as long as the error is within the fine detection range.

(23) Third, post-edge detection is introduced as a way to dramatically reduce walk error. An input signal for detection is typically amplified using a variable gain amplifier (VGA). The amplifier is designed to provide a constant output amplitude regardless of the input amplitude. First, digital programmability is introduced to the VGA to have constant delay so that it does not introduce any measurement error. Second, since the VGA has constant output amplitude, the falling edge (post-edge) of the pulse does not vary with input amplitude and hence does not suffer substantially from walk error.

(24) An architecture in accordance with many embodiments of the invention allows for multiple pulses to allow for more averaging and hence improved noise performance. Each group of pulses (at an intermediate frequency as compared to the high frequency of the fine coherent detection) can be repeated at a lower frequency hence creating yet another layer of hierarchy for multiple coherent detection. This approach enables not only fine and coarse detection but introduces an intermediate detection resolution for even larger range of detection.

(25) Pulsed-Coherent Detection

(26) In many embodiments of the invention, the received signal initiates an automatic gain control (AGC) loop when it crosses the threshold voltage ($V_{sub.th}$) of a signal detector. After the loop is settled, fine ToF is measured by coherent detection, which detects the phase difference ($\Delta\phi$) between the reference clock ($CK_{sub.ref}$) and the received signal ($RF_{sub.in}$). Instead of observing the rising edge of the envelope, it can measure the falling edge (post-edge) as coarse ToF measurement. The arrival time of the envelope can be measured by using the reference clock ($N \times T_{sub.clk}$), where N is the number of cycles (periods) and $T_{sub.clk}$ is the time duration of one cycle (period) of the reference clock. Combining measured coarse ToF and fine ToF measurement results, the total ToF can be calculated as equation (1) below:

$$(27) \quad T_oF = [N + (\frac{\text{---}}{2})] \times T_{clk} \quad (1)$$

(28) To overcome the range ambiguity, the coarse measurement has a detection precision within a clock period ($T_{sub.clk}$) and the coherent detection provides the fine ToF measurement within the cycle.

(29) A block diagram is shown in FIG. 2B to illustrate the components of a pulsed-coherent lidar system **200** in accordance with several embodiments of the invention. A variable gain analog front-end (AFE) **210** accurately controls the amplitude of the received signals. Since the amplitude is constant for a post-edge, the coarse measurement is no longer susceptible to the walk error due to varying measured rising edge of the variable amplitude inputs. This variable gain AFE (VGA) further helps reduce complexity and maintain linearity toward accurate fine detection. To measure the envelope's post-edge, the VGA output is fed to a power detector **212** prior to the counter **214**. The coherent fine phase is measured by a phase detector and an ADC **216**. The pulsed sequence can be repeated to do more averaging or be modulated by a data sequence as an identifier. In addition, the pulsed-coherent architecture can be further segmented to improve sensitivity, measurement distance, or depth resolution. Although a specific architecture is discussed with respect to FIG. 2B, one skilled in the art will recognize that any of a variety of architectures may be utilized in accordance with embodiments of the invention. Different components may be used to similar effect as discussed here.

(30) Receiver Architecture

(31) A receiver architecture **300** in accordance with several embodiments of the invention is shown in FIG. 3. An AFE provides 60-dB dynamic range, which includes 1-bit phase-invariant variable-

gain low-noise amplifier (PI-VGLNA) **312**, in-phase and quadrature-phase (I/O) down-conversion mixer **314**, 1-bit programmable gain amplifier (PGA) **316**, and a VGA **318** with a continuous AGC loop. Each amplifier provides 20-dB gain tuning range. For the digital control of the PI-VGLNA **312** and PGA **316**, a self-mixing power detector **320** after the PGA senses the input power and sets the coarse gain settings. For the continuous AGC loop, a V/I converter **322** implemented by a folded-cascode amplifier provides high gain and high output impedance. A frequency zero can be introduced in the loop filter with 250-MHz loop bandwidth to achieve fast transient response by a series R-C lowpass filter **324**. A pulse generator **326** after signal detector provides a 4-ns settling window for the AGC loop. A track-and-hold switch can hold the control-voltage of the VGAs, and release it after detecting the envelope's falling edge. In the clock generation path, a portion of the power can be split from a continuous 19-GHz source and the clock divided to provide 14.25-GHz and 4.75-GHz LOs (local oscillators) for down conversion and fine ToF detection. A superharmonic injection locked multipath ring oscillator (SHIL-MPRO) **328** and up-conversion mixers **330** can provide I/O LOs. The coarse ToF detector **332** can take the output from a self-mixing power detector **320** and a signal detector to find the envelope's post-edge transition. For fine ToF detection, two single-side band mixers (SSB Mixer) **334** can act as a coherent detector. In this architecture, an off-chip ADC **336** may be used to read out the results. While a specific architecture is described above with respect to FIG. 3, one skilled in the art will recognize that any of a variety of architectures may be utilized in accordance with embodiments of the invention. Different components may be used to similar effect as discussed here.

(32) Phase-Invariant Variable Gain Amplifiers

(33) To achieve high dynamic range, the AFE may require a wide gain tuning range and low phase error. Architectures that may be utilized for the PI-VGLNA and PGA/VGA in accordance with embodiments of the invention are shown in FIG. 4A and FIG. 4B respectively. An LNA (low noise amplifier) with current-steering cascode architecture can be chosen to stabilize the phase response of the amplifier. Moreover, the phase invariant response can be improved by inserting inductors L.sub.1 and L.sub.2 between common source and common gain stages. These inductors not only enhance input-output isolation, but they also reduce the sensitivity of phase variation. PGA/VGA can also be implemented in a current-steering structure controlled by single-to-differential V/I converters. Two flying capacitors, C.sub.1 and C.sub.2, may be used to introduce a frequency zero to compensate the phase shift at 4.75 GHz. Although specific architectures are described above with respect to FIGS. 4A and 4B, one skilled in the art will recognize that any of a variety of architectures may be utilized in accordance with embodiments of the invention. Different components may be used to similar effect as discussed here.

(34) Coarse ToF Detector

(35) A coarse ToF detector in accordance with several embodiments of the invention is shown in FIG. 5. The output of the power detector is filtered by a low-pass filter prior to counting. Since the AGC loop may have a small amount of input-dependent gain error, the filter has a small degree of adjustability to compensate for the phase shift that results. Varactors may be adjusted according to the V.sub.ctrl of the AGC loop as a signal strength indicator to adjust the transition time. The filter is followed by 8-way time-interleaved samplers, an 8-to-16 demultiplexer, and XOR gates to accurately detect the transition time at low rates. The 8-way interleaving clock phases can be provided by an SHIL-MPRO. The design, shown in FIG. 6, can be enhanced with an embedded harmonic-rejection phase interpolation in each stage to allow all phases to be shifted with a control code. The phase selector (Sel.) in FIG. 5 can then select in-phase or quadrature-phase based on the relation between I/O fine measurement to effectively double the sampling rate of the detector. This approach reduces the routing complexity of clock signals. Although specific architectures are described above with respect to FIGS. 5 and 6, one skilled in the art will recognize that any of a variety of architectures may be utilized in accordance with embodiments of the invention. Different components may be used to similar effect as discussed here.

(36) Experimental Results

(37) A pulsed-coherent detector in accordance with an embodiment of the invention has been designed and fabricated in 28-nm CMOS technology. The receiver consumes 121 mA from 1-V supply. A measurement setup is illustrated in FIG. 7. The performance results presented here are prior to integration with the optical components and hence use the electrical interface. The transmitter is built by a synthesizer, a pulse generator and an up-conversion mixer to generate pulse-modulated signals. The receiver performance was characterized by tuning an attenuator and a phase shifter in the signal path. An off-chip ADC captures the outputs from coherent detection or direct samples from 4.75-GHz signals for characterizing the response of AGC loop.

(38) FIG. 8 shows the measurement results of the phase-invariant AFE. The phase error across 20-dB tuning range is $\pm 0.14^\circ$ of PI-VGLNA, $\pm 0.11^\circ$ of PGA, and $\pm 0.5^\circ$ of VGA, which corresponds to $\pm 16.5 \mu\text{m}$ resolution accuracy. FIGS. 9A and 9B depict the AGC loop characteristic. Loop gain error of 20-dB input power difference is shown in FIG. 9A, the gain error is within 0.3 dB. The transient response of received pulsed modulated waveform is captured and shown in FIG. 9B with setting time less than 4 ns. FIG. 9C and FIG. 9D compare the walk error between rising edge detection and the proposed falling edge detection with different received input powers. The relative input power is referenced to the minimum detectable SNR of 12 dB before the signal detector. For a single-threshold rising-edge detection, the walk error across 16-dB dynamic range can be as large as 600 ps (FIG. 9C). The proposed post-edge detection reduces the walk error to 26 ps (FIG. 9D). With the error less than one cycle, the coherent detection results are used and the error no longer impacts the accuracy of the measured ToF.

(39) FIG. 10A depicts the rms error (e) of the coarse ToF readouts with 70 detected post-edges, which can be given as equation (2) below:

$$(40) \quad e = \frac{1}{\sqrt{N}} \quad (2)$$

(41) The rms error of the fine ToF measurement is shown in FIG. 10B. With 1- μs integration time ($N=140$), the rms error is 130 fs with an SNR of 12 dB and 42 fs with high SNR. The corresponding precision is 19.5 μm and 6.3 μm respectively. Combining the rms error and the phase offset due to the VGAs, the precision of 40 μm can be achieved based on the timing accuracy. FIG. 11 shows the spectrum of the received pulse sequence at the VGA's output, and FIG. 12 shows the phase noise of 4.75-GHz clock for coherent measurement. The rms jitter integrated from 1 kHz to 40 MHz is 100 fs. Table I summarizes the performance detectors in accordance with several embodiments of the invention.

(42) This design presents a detection architecture using pulsed-coherent coarse-fine ToF measurement that combines the benefits of pulsed detection and coherent detection in a segmented converter design. Post-edge envelope detection reduces the sensitivity to walk error so that the pulsed detection is sufficiently accurate as the coarse measurement for the fine coherent detection. The fast settling AGC loop enhances the sampling rate, and the phase-invariant variable gain analog front-end improves the precision. This ToF receiver can theoretical achieve 40- μm resolution with 1-MHz sampling rate.

(43) Processes for Measuring Distance Using Both Coherent and Pulsed ToF Detection

(44) Processes for measuring distance with a light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF) in accordance with embodiments of the invention may utilize hardware such as those described further above. A process 1300 for measuring distance using a lidar system in accordance with several embodiments of the invention is shown in FIG. 13.

(45) The process 1300 includes providing or receiving (1302) a clock signal (CK.sub.ref) with time period T.sub.clk from a reference clock. An automatic gain control (AGC) loop is triggered (1304) when a received signal RF.sub.in is greater than a threshold voltage V.sub.th. The process includes measuring (1306) a fine ToF using a coherent detector by detecting the phase difference ($\Delta\phi$)

between the clock signal (CK.sub.ref) and the received signal (RF.sub.in). The process also measures (1308) a coarse ToF using a pulse edge detector by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$. The total ToF is calculated (1310) by combining output of the coherent detector (fine ToF) and pulse edge detector (coarse ToF), such as by using the equation:

$$(46) \text{ } T o F = [N + (\frac{\text{---}}{2})] \times T_{\text{clk}} .$$

(47) Although a specific process is described above with respect to FIG. 13, one skilled in the art will recognize that any of a variety of processes may be utilized for ToF measurement including coherent and pulsed detection in accordance with embodiments of the invention.

(48) Although the description above contains many specificities, these should not be construed as limiting the scope of the invention but as merely providing illustrations of some of the presently preferred embodiments of the invention. Various other embodiments are possible within its scope. Accordingly, the scope of the invention should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

Claims

1. A light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF) measurement, comprising: a reference clock providing a clock signal (CK.sub.ref) with time period $T_{\text{sub.clk}}$; an automatic gain control (AGC) loop that is triggered when a received signal RF.sub.in is greater than a threshold voltage $V_{\text{sub.th}}$; a coherent detector configured to measure a fine ToF by detecting the phase difference ($\Delta\phi$) between the clock signal (CK.sub.ref) and the received signal (RF.sub.in); a pulse edge detector configured to measure a coarse ToF by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$; and a combiner configured to calculate total ToF by combining output of the coherent detector and pulse edge detector using the equation:

$$T o F = [N + (\frac{\text{---}}{2})] \times T_{\text{clk}} .$$

2. The lidar system of claim 1, wherein the AGC loop comprises a folded-cascode amplifier as a V/I converter.

3. The lidar system of claim 1, wherein the coherent detector comprises two single-side band (SSB) mixers.

4. The lidar system of claim 1, further comprising a variable gain analog front-end to control amplitude of the received signal (RF.sub.in).

5. The lidar system of claim 4, wherein the variable gain analog front-end comprises a phase-invariant variable-gain low-noise amplifier (PI-VGLNA), in-phase and quadrature phase (I/Q) down-conversion mixer, programmable gain amplifier (PGA) and variable gain amplifier (VGA).

6. The lidar system of claim 5, wherein the PI-VGLNA comprises a current-steering cascode architecture with inductors between common source and common gain stages.

7. The lidar system of claim 5, wherein the PGA and VGA comprise current-steering structures controlled by single-to-differential V/I converters.

8. The lidar system of claim 1, wherein the pulse edge detector comprises varactors adjusted according to voltage $V_{\text{sub.ctrl}}$ of the AGC loop, 8-way time-interleaved samplers, an 8-to-16 demultiplexer, and XOR gates.

9. A method for measuring distance with a light detection and ranging (lidar) system utilizing both coherent and pulsed detection for Time of Flight (ToF), the method comprising: providing a clock signal (CK.sub.ref) with time period $T_{\text{sub.clk}}$ from a reference clock; triggering an automatic gain control (AGC) loop when a received signal RF.sub.in is greater than a threshold voltage $V_{\text{sub.th}}$; measuring a fine ToF using a coherent detector by detecting the phase difference ($\Delta\phi$) between the clock signal (CK.sub.ref) and the received signal (RF.sub.in); measuring a coarse ToF using a pulse

edge detector by detecting a falling edge (post-edge) of the received signal (RF.sub.in) and counting cycles N to estimate an arrival time of $N \times T_{\text{sub.clk}}$; and calculating total ToF by combining output of the coherent detector and pulse edge detector using the equation:

$$T_{\text{of}} = [N + (\frac{1}{2})] \times T_{\text{clk}} .$$

10. The method of claim 9, wherein the AGC loop comprises a folded-cascode amplifier as a V/I converter.
 11. The method of claim 9, wherein the coherent detector comprises two single-side band (SSB) mixers.
 12. The method of claim 9, further comprising controlling amplitude of the received signal (RF.sub.in) using a variable gain analog front-end.
 13. The method of claim 12, wherein the variable gain analog front-end comprises a phase-invariant variable-gain low-noise amplifier (PI-VGLNA), in-phase and quadrature phase (I/Q) down-conversion mixer, programmable gain amplifier (PGA) and variable gain amplifier (VGA).
 14. The method of claim 13, wherein the PI-VGLNA comprises a current-steering cascode architecture with inductors between common source and common gain stages.
 15. The method of claim 13, wherein the PGA and VGA comprise current-steering structures controlled by single-to-differential V/I converters.
 16. The method of claim 9, wherein the pulse edge detector comprises varactors adjusted according to voltage V.sub.ctrl of the AGC loop, 8-way time-interleaved samplers, an 8-to-16 demultiplexer, and XOR gates.
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